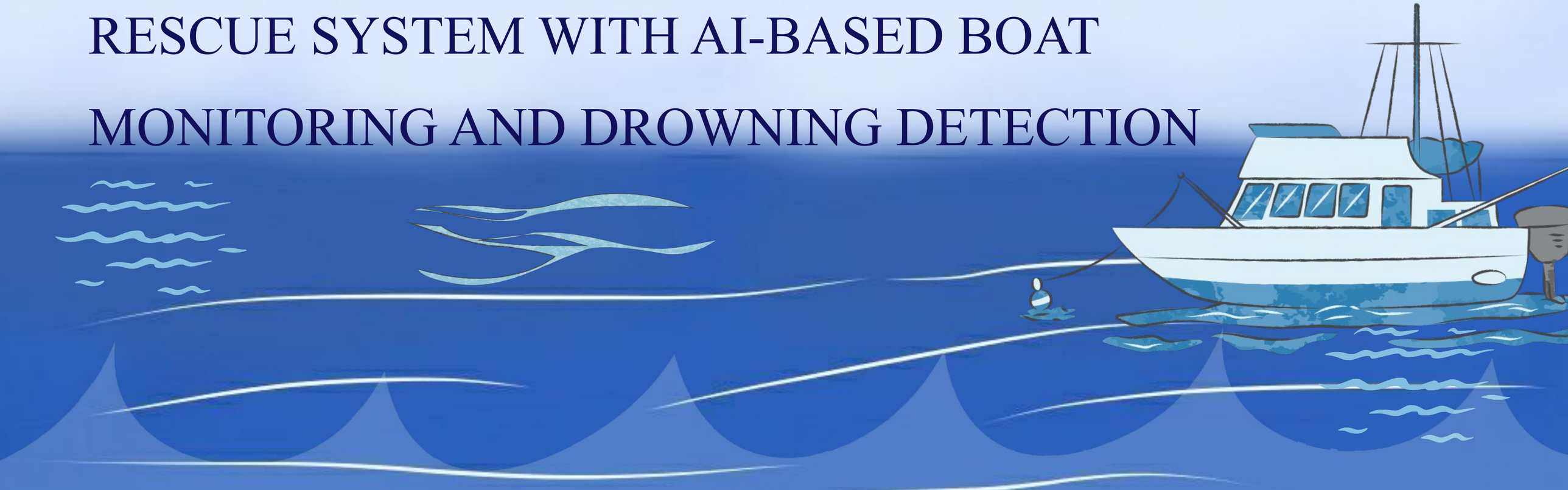
SMART FISHERIES SURVEILLANCE AND RESCUE SYSTEM WITH AI-BASED BOAT MONITORING AND DROWNING DETECTION



Introduction

Illegal fishing during trolling ban periods and delayed detection of drowning incidents are serious issues affecting marine ecosystemum human life-um. Existing systems mostly rely on manual monitoring, which is inefficient and time-consuming.

This project, Smart Fisheries Surveillance and Rescue System with AI-based Boat Monitoring and Drowning Detection, uses Artificial Intelligence (AI) and Computer Vision to provide automated coastal surveillance and fast rescue assistance.

Objectives

- Detect fishing boats during trolling ban periods automatically
- Identify drowning persons using AI-based video analysis
- Send real-time alerts to authorities, rescue teams, and boat users
- Enable fast and coordinated rescue operations

Problem Statement

Existing fisheries monitoring systems rely on manual patrols and delayed reporting, leading to poor enforcement of fishing regulations and slow rescue responses. There is a need for an automated system that can monitor coastal areas continuously, detect violations and emergencies in real time, and support quick rescue actions.

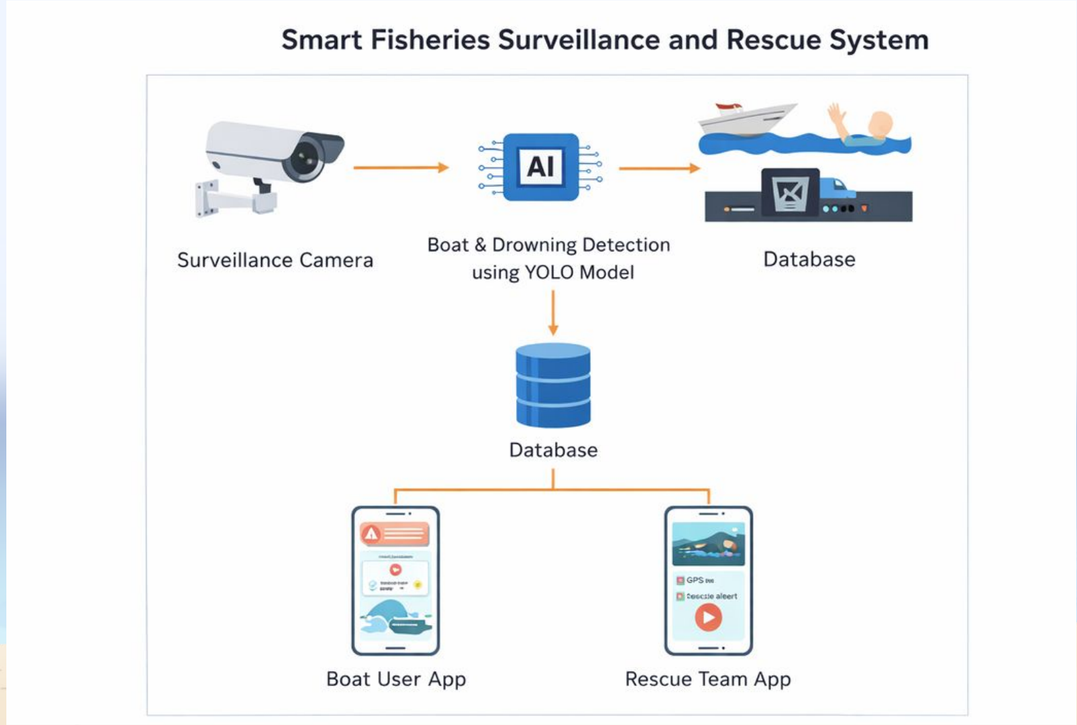
Methodology

The system uses surveillance cameras to capture live video feeds, which are analyzed using AI-based computer vision models to detect boats and drowning persons. Detected events are validated against predefined rules and instantly shared with concerned users through web and mobile applications, enabling rapid response and effective monitoring.

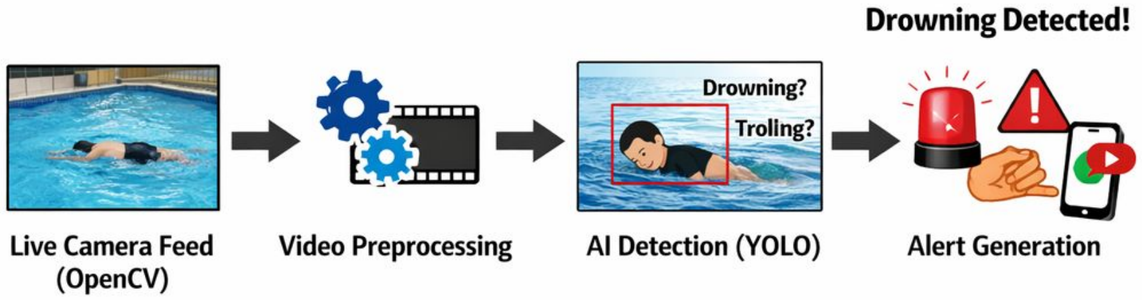
Technology Stack

Layer	Technology
AI & CV	Python, OpenCV, YOLO (v8), TensorFlow/PyTorch
Backend	Django (REST API), MySQL
Frontend Web	HTML, CSS, JavaScript, Bootstrap
Mobile Apps	Flutter (Dart), Google Maps API, Firebase
Real-Time Alerts	Firebase Cloud Messaging, Twilio, WebSockets
Edge Processing	NVIDIA Jetson / Raspberry Pi
Development IDEs	PyCharm, VS Code, Android Studio
Hardware	Core i3+, 8GB RAM+, 40GB+ HDD, Camera sensors, GPS-enabled devices

System Architecture



Process Flow Diagram



Modular Design

1. Admin Module

- User & Camera Management: Add, remove, and manage fishermen, rescue teams, and surveillance cameras.
- Trolling Ban Management: Define ban zones, schedules, and rules.
- Alert Monitoring: View and manage alerts for unauthorized boats or drowning incidents.
- Reports & Logs: Access compliance reports, camera logs, and rescue operation history.

Modular Design

1. Boat User (Fisherman) Module

- Registration & Profile Management: Create and update account information.
- Trolling Ban Alerts: Receive real-time notifications if entering restricted zones.
- Weather & Safety Updates: Check weather conditions, sea alerts, and safety notifications.
- Emergency Rescue Request: Send distress alerts with live GPS location.
- Optional: Voice or vibration alerts at sea.

Modular Design

3. Rescue Team Module

- Alert Reception: Receive real-time alerts for drowning incidents.
- Incident Tracking: View GPS location, snapshots, and video feeds.
- Live Feed Preview: Monitor incidents in real-time through camera streams.
- Incident Management: Mark incidents as resolved or in progress.

Sprint 1

Backlog Term	Date	Original Estimation (hrs)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Project setup & environment configuration	15/01/26	3	1	1	1	0	0	0	0
Database schema design (users, boats, alerts, AI logs)	16/01/26	4	1	1	1	1	0	0	0
Admin Module (user & camera management, ban schedules)	18/01/26	5	1	1	1	1	1	0	0
Total	–	**12**	3	3	3	2	1	0	0

Sprint 2

Backlog Term	Date	Original Estimation (hrs)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
AI Surveillance Module (boat detection)	20/01/26	3	1	1	1	0	0	0	0
Drowning person detection	21/01/26	3	1	1	1	0	0	0	0
Video preprocessing	22/01/26	2	1	1	0	0	0	0	0
Alert generation (AI → Backend)	23/01/26	2	1	1	0	0	0	0	0
Total	–	**10**	4	4	2	0	0	0	0

Sprint 3

Backlog Term	Date	Original Estimation (hrs)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Boat User Mobile App (alerts, rescue request)	23/01/26	5	1	1	1	1	1	0	0
Rescue Team Mobile App (alert reception, live feed)	25/01/26	5	1	1	1	1	1	0	0
Frontend Web Portal (dashboard, reports)	27/01/26	5	1	1	1	1	1	0	0
Integration & Testing (all modules together)	31/01/26	6	1	1	1	1	1	1	0
Total	–	**21**	4	4	4	4	4	1	0

Thank You